IDEA Classroom Computer: Installation Manual

Raspberry Pi 400 + Raspberry Pi Monitor + Ubuntu Desktop

For Justice. Version 1.2, 15 October 2026

IDEA - offline school computing. Prepared for Koen Swings's team.
No real passwords, keys or tokens are included; values in <ANGLE BRACKETS> are placeholders.

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How to use this manual

This manual sets up one classroom computer: a Raspberry Pi 400 keyboard computer with the official 15.6" Raspberry Pi Monitor, running Ubuntu Desktop. The computer opens the school's IDEA Console. Each step has:

1. **Do:** what to do.
2. ✔ **Check:** how to see that it worked.

Placeholders, no real secrets.

- Every school uses the same addresses (*IDEA School Server: Installation Manual*). The Console is at `http://idea.local`, whichever and however many Engines the school has.
- `<STUDENT-USER-PASSWORD>`: the password for the Ubuntu user on the Pi 400.
- `<SCHOOL>`: the school name, for labels only.

1. What is in the boxes

1.1 Raspberry Pi 400 kit (bol.com, "Raspberry Pi 400 kit, muis, 16GB microSD, English guide, QWERTY")

> The bol.com page could not be opened from our tools (bol.com blocks automated access). The list below is the **official Raspberry Pi 400 kit** contents from raspberrypi.com [1]. Check it against the box.

☐	Item	Use it?
☐	Raspberry Pi 400 (English QWERTY keyboard, 4 GB)	Yes
☐	Official 15 W USB-C power supply (5.1 V / 3 A)	Yes: it powers the Pi **and** the monitor
☐	Raspberry Pi USB mouse	Yes
☐	16 GB microSD card with Raspberry Pi OS on it	**Keep it** as a rescue card (EEPROM update, 6.1). Ubuntu goes on Koen's **32 GB** cards.
☐	micro-HDMI to HDMI cable	Yes: Pi 400 → monitor
☐	Beginner's Guide book	Optional

Check the **plug** of the power supply. Kits sold in Belgium usually have the European (Type C/E/F) plug. Zimbabwe uses Type G (UK-style) and Type D sockets, so bring a good adapter for each computer, or a power strip that takes EU plugs.

1.2 Raspberry Pi Monitor 15.6" red/white (Kiwi Electronics, SC0940)

From the Kiwi product page [2] and the Raspberry Pi docs [3]:

☐	Item
☐	Monitor: 15.6", 1920 × 1080 IPS, built-in stand, 2 × 1.2 W speakers
☐	**USB-A to USB-C cable** (for power). This is the only cable in the box: there is **no HDMI cable** and **no power supply**.

Monitor connectors: **HDMI** in (standard size, HDMI 1.4), **USB-C** (power in only), **3.5 mm** audio out. On the back are buttons for standby, brightness up/down and volume up/down.

1.3 From Koen

☐	Item
☐	One **32 GB microSD card** for each Pi 400 (fast, A1/A2 class is best). Use the same brand/batch if you want to clone cards (5.4).
☐	Stickers or a marker for labels: school name, computer number

2. Power: one power supply for the Pi 400 and the monitor

Koen wants **one** power supply per workstation. That works. Here is what the Raspberry Pi documentation says:

- **The monitor can be powered from the Pi.** "If you're using an official Raspberry Pi Monitor, you can connect your monitor to power through a USB cable connected to your Raspberry Pi keyboard computer. When you power on your keyboard computer, it then provides power to the Raspberry Pi Monitor." [1]
- **But only in Power Saving mode.** "When powered from a Raspberry Pi, the Monitor operates in Power Saving mode, since Raspberry Pi devices cannot provide 1.5 A of current over a USB-A connection." In Power Saving mode, brightness is limited to **0-50 %** and speaker volume to **0-60 %**. [3]
- **Full Power mode** (100 % brightness and volume) needs a USB-PD source with **at least 1.5 A** [3]. A Pi 400 can give its USB ports **1.2 A in total** [4].
- The monitor uses **4.5-6.5 W** while on [3], which is about 0.9-1.3 A at 5 V.

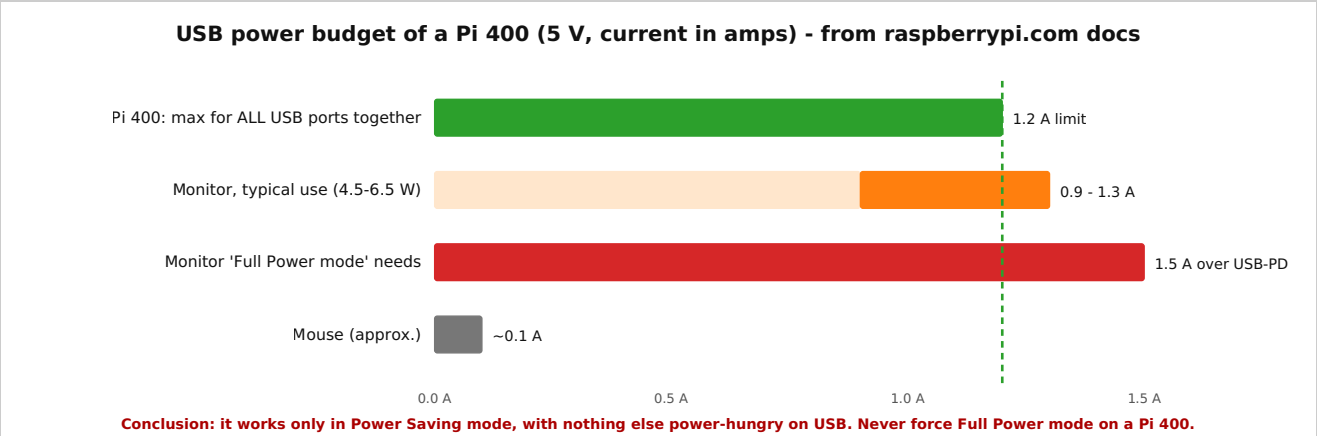


Figure 1. USB power budget

Rules for the one-power-supply setup

1. Use the **official 15 W (5.1 V / 3 A)** Raspberry Pi USB-C supply from the kit [5]. Never use a phone charger.
2. Plug **nothing else power-hungry** into the Pi 400's USB ports: only the mouse (and at most a USB stick). The monitor and mouse together are already close to the 1.2 A limit.
3. **Never force Full Power mode** on a monitor powered by a Pi 400. That is done by holding brightness-up for 3 seconds [3]. It would ask for more current than the Pi can give.
4. If the monitor **flashes on and off**, or the Pi shows a lightning-bolt/under-voltage warning, the supply can't keep up. Turn brightness and volume down. If that doesn't help, give the monitor its own USB-C power supply (a second official 15 W supply works) [3].

2.1 Wiring

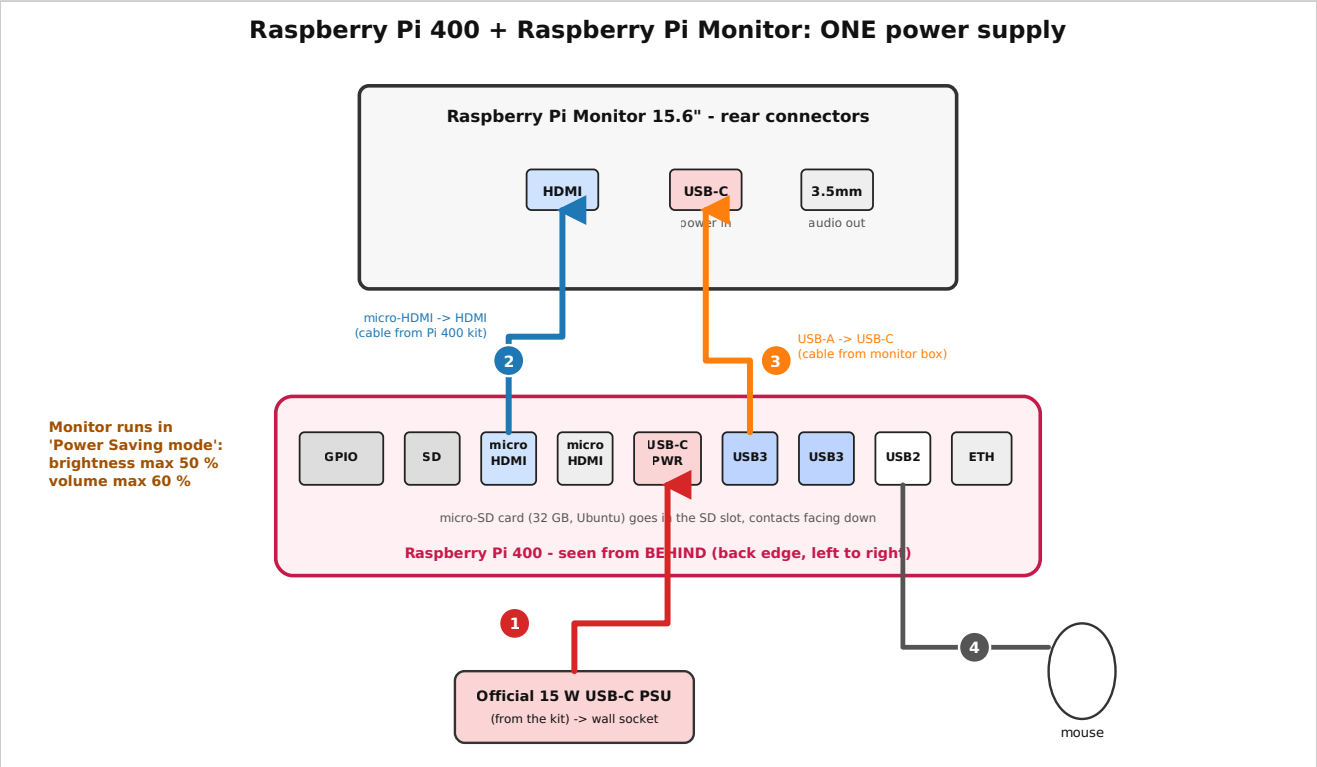


Figure 2. Pi 400 and monitor wiring

Cable	From	To
1	Official 15 W PSU (wall socket)	Pi 400 **USB-C power** port
2	Pi 400 **micro-HDMI** port, the one next to the microSD slot	Monitor **HDMI** (cable from the Pi 400 kit)
3	Pi 400 **USB 3.0** port (blue)	Monitor **USB-C** (USB-A → USB-C cable from the monitor box)
4	Mouse	Pi 400 **USB 2.0** port (white). Raspberry Pi recommends the white port for the mouse [1].

The back edge of the Pi 400, seen from **behind**, left to right: GPIO, microSD, micro-HDMI, micro-HDMI, USB-C power, USB 3.0, USB 3.0, USB 2.0, Ethernet. Small icons are printed next to each port. If you get no picture, move cable 2 to the other micro-HDMI port.

Do (in this order):

1. Put the 32 GB microSD card in the slot, contacts facing down (only after section 5).
2. Connect cable 2 (HDMI), cable 3 (monitor power) and cable 4 (mouse).
3. Connect cable 1 (power) **last**. The Pi 400 starts at once. Later you switch it on and off with **Fn + F10** [1].

✓ Check:

- The Pi 400's LED lights up. Within 10-20 seconds the monitor shows a picture (rainbow/boot screen or Ubuntu logo).

- Press the monitor's brightness-up button: the on-screen bar stops at 50. That is Power Saving mode, as expected.
- The picture stays stable for 5 minutes with no flashing.

3. Which Ubuntu?

Release	Status on 12 Oct 2026	Pi 400 support	Choice
Ubuntu 26.04.1 LTS "Resolute Raccoon" (Desktop, 64-bit)	Latest **LTS**, 5 years of updates. Listed in Raspberry Pi Imager (released 27 Aug 2026).	**Desktop supported** on Pi 400 (Ubuntu's support table: "SD") [6]	**Use this one**
Ubuntu 26.10 "Stonking Stingray"	Final release planned for **15 Oct 2026** [7]. An interim release, 9 months of support.	Probably, but not yet released	Not for schools
Ubuntu 24.04.5 LTS	Older LTS	Supported	Only as a fallback

Ubuntu marks the Pi 400 as **supported** for Desktop, but **not certified** (certification for 26.04 covers the Pi 4B and Pi 5) [6]. The Pi 400 has 4 GB of RAM, which is enough for a browser-based Console, but don't expect it to be fast.

> **Minimum bootloader (EEPROM) date.** Ubuntu 24.04 and 26.04 need a Pi 400 EEPROM dated **2022-11-25 or later** [6]. New kits almost always meet this. If a Pi 400 won't boot Ubuntu, update the EEPROM with the kit's 16 GB Raspberry Pi OS card (6.1).

4. Install Raspberry Pi Imager (on Ubuntu, Mac or Windows)

Your computer	How
Ubuntu (also Ubuntu on a Pi)	`sudo snap install rpi-imager` [6], or `sudo apt install rpi-imager`
macOS / Windows	Download it from `https://www.raspberrypi.com/software/` and install it

The current Imager is **2.0.12** (8 Oct 2026).

✔ **Check:** Imager starts and shows the steps Device → OS → Storage.

> **Imager cannot pre-set the hostname, user, Wi-Fi or locale for Ubuntu Desktop.** In Imager's own OS list, "Ubuntu Desktop 26.04.1 (64-bit)" has `init_format: none`, so Imager offers **no OS customisation** for it. That only works for Raspberry Pi OS and Ubuntu Server. Ubuntu Desktop asks for these settings in its **first-boot wizard** (section 6) [6][8]. This manual sets them there.

5. Write the microSD cards

5.1 Download the image once (saves time and data)

Download it once on a good connection in Belgium:

- URL: `https://cdimage.ubuntu.com/releases/26.04.1/release/ubuntu-26.04.1-preinstalled-desktop-arm64+raspi.img.xz`
- Size: about 3.1 GB. It unpacks to 9.6 GB, so it fits a 32 GB card.

✔ **Check:** `sha256sum ubuntu-26.04.1-preinstalled-desktop-arm64+raspi.img.xz` prints `8216f54cc1df2fcf7de37530a7edccdd200abd7dcfbc5d4f0407df7513544d6a`. This value is from Ubuntu's `SHA256SUMS` and from Imager's OS list.

5.2 Write one card with Imager

1. Put a 32 GB card in your card reader.
2. Imager → **Device:** Raspberry Pi 4 (this covers "Pi 4 Model B, **400**, and CM4").
3. **OS:** Use custom → pick the downloaded `.img.xz`. Or choose Other general-purpose OS → Ubuntu → Ubuntu Desktop 26.04.1 (64-bit), but then Imager downloads it again.
4. **Storage:** the 32 GB card. Check the size so that you don't overwrite your own disk!
5. Imager offers no customisation for this image (see the box above). Click **Write** and confirm that the card will be erased.
6. Wait for **writing and verification** to finish (about 10-15 minutes), then remove the card.

✔ **Check:** Imager says Write successful. Put a label on the card: `<SCHOOL> - PC __`.

5.3 Batch tips for many cards

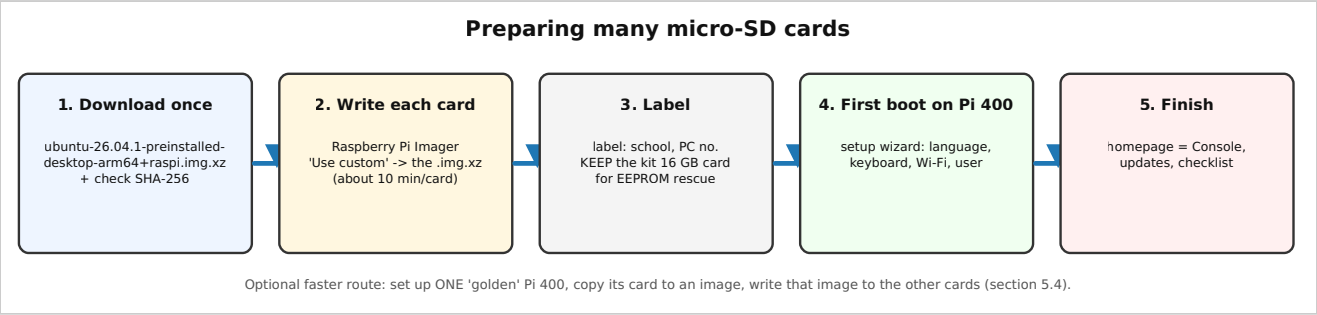


Figure 3. Card preparation

- Use the **same downloaded** `.img.xz` for every card ("Use custom"), not a fresh download each time.
- Imager writes one card at a time. With 2-3 USB card readers you can run a second Imager window on a second reader. Watch the **Storage** choice every time.
- On Linux you can also write from the command line with `rpi-imager --cli <image.img.xz> /dev/sdX`. Check `/dev/sdX` with `lsblk` first: a wrong letter erases the wrong disk. **Test first** on one card.

- Keep the kit's **16 GB Raspberry Pi OS cards** in a labelled bag. They are your EEPROM-rescue cards.

5.4 Faster: one "golden" card, then copies

Fresh cards each need the first-boot wizard (about 10 minutes per Pi). A golden card lets you do the wizard, the updates and the homepage **once**:

1. Set up one Pi 400 fully, sections 6.1 to 6.6, on your **home Wi-Fi**, with all updates.
2. Before you shut it down, make the copy "anonymous" so that each clone gets its own machine ID:

```
sudo truncate -s 0 /etc/machine-id
sudo rm -f /var/lib/dbus/machine-id
sudo shutdown -h now
```

3. On your Ubuntu machine, read the card into an image, then shrink and compress it:

```
lsblk                                  # find the card, e.g. /dev/sdb
sudo dd if=/dev/sdb of=golden-pi400.img bs=4M status=progress conv=fsync
# optional: PiShrink (github.com/Drewsif/PiShrink) so it fits cards that are a few MB smaller
xz -T0 -v golden-pi400.img
```

4. Write `golden-pi400.img.xz` to each card with Imager → Use custom, as in 5.2.
5. On each clone, give it its own name after the first boot: `sudo hostnamectl set-hostname idea-pc07` (PC 07 of this school).

Test first on two cards before you make 20. Check that both clones boot, each have a different `cat /etc/machine-id`, and both reach the Console. Without PiShrink, the target card must be at least as large as the golden card. Two "32 GB" cards of different brands often differ by a few MB.

6. First boot and setup on the Pi 400

6.1 (Only if it won't boot) Update the EEPROM with the kit card

1. Put the kit's **16 GB Raspberry Pi OS** card in the Pi 400 and boot it. Its own setup wizard will appear: complete it with any user.
2. Open a terminal and run `sudo rpi-eeprom-update`. It shows the current bootloader date.
3. If the date is older than 2022-11-25, run `sudo rpi-eeprom-update -a`, then `sudo reboot` (the reboot does the flashing). The Ubuntu documentation describes the same procedure from an Ubuntu 24.04 image [6].

✔ **Check:** `sudo rpi-eeprom-update` now shows a date of 2022-11-25 or later. Swap back to the Ubuntu card.

6.2 The first-boot wizard

1. Put the Ubuntu card in. Connect power. First boot takes a few minutes: the card's partition is grown and services are set up.
2. The wizard asks for these settings. The exact screens can differ a little between Ubuntu versions: * **Language:** English * **Keyboard:** English (US). The kit is a QWERTY "English" keyboard. * **Wi-Fi:** see 6.3 * **Location / time zone:** Harare (Africa/Harare) * **User:** for example name `Student`, username `student`, password `<STUDENT-USER-PASSWORD>`. Make one shared user per PC. Do **not** tick log in automatically: the class teacher gives students the password. * Skip Ubuntu Pro / online accounts / location services.
3. The computer name: if the wizard doesn't ask, set it afterwards in **Settings → System → About → Device Name**, for example `idea-pc07`.

✔ **Check:** you reach the Ubuntu desktop, and **Settings → About** shows Raspberry Pi 400, Ubuntu 26.04.1 LTS and your device name.

6.3 Wi-Fi: IDEA or IDEA-Teachers

The pre-flashed IDEA school router creates two Wi-Fi networks (IDEA School Server: Installation Manual, "The school router's Wi-Fi networks and default passwords"). The passwords are the IDEA defaults until the school changes them:

Who uses this PC	Network
Students (default)	`IDEA`, password `idea911!`: Console and apps only, no internet
Teacher's PC	`IDEA-Teachers`, password `idea-teachers911!`: Console and apps, plus internet when the teacher rule is on (off by default)

Use `IDEA` on every PC unless the head teacher names a PC for a teacher. Only that PC gets `IDEA-Teachers`. Never type the `IDEA-Teachers` password on a student PC.

If the school has changed the passwords (IDEA School Server: Installation Manual, section 10), ask the head teacher for the new ones.

At school: choose the network in the wizard, or later from the top-right menu → Wi-Fi.

Preparing in Belgium: the IDEA network isn't there yet. Join your **home Wi-Fi** for updates (6.5), then add IDEA in advance from a terminal.

```
nmcli connection add type wifi con-name IDEA ssid IDEA \
  wifi-sec.key-mgmt wpa-psk wifi-sec.psk 'idea911!' connection.autoconnect yes
# when done with updates, forget your home network:
nmcli connection delete '<HOME-WIFI-NAME>'
```

Wi-Fi country (recommended). Ubuntu Desktop on the Pi doesn't keep a Wi-Fi country setting, and some 5 GHz channels need it [8]. Add `cfg80211.ieee80211_regdom=ZW` to the **end of the single line** in `/boot/firmware/cmdline.txt`:

```
sudo sed -i '1 s/$/ cfg80211.ieee80211_regdom=ZW/' /boot/firmware/cmdline.txt
cat /boot/firmware/cmdline.txt     # still ONE line, ending in ...regdom=ZW
```

✔ **Check:** at school, the Wi-Fi icon shows "IDEA" and `ip -4 addr show wlan0` shows `10.42.1.x`. Ubuntu may say "limited connectivity" or show no internet: that is **normal** on IDEA. After a reboot, `iw reg get` shows `country ZW`.

6.4 Homepage = the school Console

The homepage is the school Console at **`http://idea.local`**. The port is 80 (the IDEA Engine's `config.yaml httpPort: 80`), so no `:port` is needed. A school has one or more Engines (1 to 4 today), and which address each one gets is not fixed. The IDEA router answers `idea.local` with the Engines that are switched on, so the homepage works with one Engine or four, and keeps working when one is off. Do not use a sticker name here: the PCs are on the Wi-Fi network, and sticker `.local` names do not cross from the wired network to Wi-Fi.

The Console reads the list of all Engines from the IDEA database, so every Engine appears on this one page. If `idea.local` ever fails, type an Engine's address by hand (`http://10.42.0.2` to `.9`; the school's fill-in sheet lists them).

Firefox (Ubuntu's default browser) for all users, set once by the admin. The Firefox snap reads enterprise policies from `/etc/firefox/policies/policies.json`:

```
sudo mkdir -p /etc/firefox/policies
sudo tee /etc/firefox/policies/policies.json > /dev/null <<'JSON'
{
  "policies": {
    "Homepage": { "URL": "http://idea.local/", "StartPage": "homepage", "Locked": false },
    "OverrideFirstRunPage": "",
    "OverridePostUpdatePage": "",
    "DisableTelemetry": true
  }
}
JSON
```

(The address is the same at every school. Restart Firefox.)

Or by hand, in Firefox: ☰ → **Settings → Home** → Homepage and new windows → **Custom URLs** → `http://idea.local/`.

Chromium (only if you install it: `sudo snap install chromium`): ⋮ → **Settings → On start-up → Open a specific page** → `http://idea.local/`.

Handy: put Firefox in the dock (right-click → Pin to Dash). Open the Console, then ★ to bookmark it.

✔ **Check:** close and reopen Firefox. The IDEA Console opens straight away, and **about:policies** shows the policy as Active.

6.5 Updates (in Belgium, on home Wi-Fi)

The IDEA Wi-Fi has **no internet**, so update before you leave:

```
sudo apt update && sudo apt full-upgrade -y
sudo snap refresh
sudo reboot
```

✔ **Check:** `sudo apt update` ends with "All packages are up to date".

6.6 Final touches

- Power off with **Fn + F10** or the system menu → Power Off. Wait until the screen is black before you pull the plug.
- The monitor's speakers work over HDMI. If there is no sound, choose HDMI under **Settings → Sound → Output**.

7. Checklist for each computer

☐	Test	Expected
☐	Wiring as in 2.1, power plugged in last	Picture within about 20 s, no flashing
☐	Monitor brightness up	Bar stops at 50 (Power Saving mode, expected)
☐	Ubuntu desktop reached, device name set	Settings → About: Raspberry Pi 400, Ubuntu 26.04.1 LTS
☐	Wi-Fi	Connected to `IDEA` (or `IDEA-Teachers`), address `10.42.1.x` / `10.42.2.x`
☐	Wi-Fi country	`iw reg get` → `country ZW`
☐	Firefox opens the Console	`http://idea.local` loads and lists **every** Engine of the school
☐	Apps open from the Console	Kolibri / Kiwix / Nextcloud open
☐	No internet on IDEA	`https://example.com` fails (expected)
☐	Power stable	No lightning-bolt or under-voltage warning during 10 minutes of use
☐	Label	Card and PC labelled `<SCHOOL> - PC __`

Appendix A. Troubleshooting

Problem	Try
No picture	Move cable 2 to the other micro-HDMI port. Check that the monitor isn't in standby (press ⏻).
Monitor flashes on and off	Turn brightness and volume down [3]. Unplug extra USB devices. Last resort: a separate USB-C supply for the monitor.
Rainbow screen / no boot with Ubuntu	EEPROM too old: section 6.1. Rewrite the card and check the SHA-256.
Very slow first boot	That is normal: the first boot resizes the card. Give it 5 minutes.
Wi-Fi 'IDEA' not shown	Is the router on? (IDEA School Server: Installation Manual, section 2.) The 5 GHz network may need the Wi-Fi country (6.3).
Console doesn't open	Check the address: `http://idea.local` (http, not https). Is the PC on IDEA/IDEA-Teachers? Run `ping idea.local`; if that fails, try an Engine address from the fill-in sheet.

Appendix B. Sources (read on 12 October 2026)

1. Raspberry Pi documentation, Keyboard computers (Pi 400 kit contents, setup, monitor powered from the keyboard, Fn + F10): `raspberrypi.com/documentation/computers/keyboard-computers.html`.
2. Kiwi Electronics, Raspberry Pi Monitor 15.6" Rood/Wit (KW-3393, SC0940; package: monitor + USB-A to USB-C cable): `https://www.kiwi-electronics.com/nl/raspberry-pi-monitor-15-6quot-rood-wit-20221`.
3. Raspberry Pi documentation, Raspberry Pi Monitor (Power the Monitor, Power Saving mode 0-50 % brightness / 0-60 % volume, 1.5 A USB-PD, 4.5-6.5 W): `raspberrypi.com/documentation/accessories/monitor.html`.
4. Raspberry Pi documentation, Power supply table (Pi 400: 3.0 A PSU, 1.2 A total USB peripheral current): `raspberrypi.com/documentation/computers/raspberry-pi.html#power-supply`.
5. Raspberry Pi documentation: "For Raspberry Pi 4 and Raspberry Pi 400, we recommend the 3A USB-C Supply".
6. Ubuntu, Install Ubuntu on the Raspberry Pi (support table, minimum EEPROM dates, desktop image first-time setup wizard): `https://ubuntu.com/hardware/docs/boards/how-to/ubuntu_supported/raspberry-pi/`, and `ubuntu.com/download/raspberry-pi`.
7. Ubuntu 26.10 release schedule: `https://documentation.ubuntu.com/release-notes/26.10/schedule/`.
8. Raspberry Pi Imager OS list `os_list_imagingutility_v4.json` (Ubuntu Desktop 26.04.1: `init_format: none`). rpi-imager issue #862 (Ubuntu Desktop customisation, Wi-Fi regulatory domain via `cfg80211.ieee80211_regdom`).